P425/1
MATHEMATICS
Paper 1
29 July 2026
3 hours

ENTEBBE

ENTEBBE JOINT EXAMINATION BUREAU

Uganda Advanced Certificate of Education

MATHEMATICS

Paper 1

3 hours

INSTRUCTIONS TO CANDIDATES:

The paper has **two** Sections*:* **A** *and* **B**. *It consists of* **five** *items.*

Section ***A*** *has* **one** *compulsory item.*

Section ***B*** *has* **two** *parts:* I *and* II, *Respond* to ***one*** *item from each* Part.

Any additional item(s) responded to will **not** *be scored.*

All responses must be written in the answer booklet(s) provided.

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

SECTION A

Items in this Section *are compulsory*

ITEM 1

A construction company is designing two circular recreational parks within a new housing estate. Before construction begins, the engineering team must determine the exact boundary of each circular park using three survey markers placed on the ground.

Park A: Three survey markers are located at *A(0,1)m, B(4,3)m***,** and *C(1,-1)m*.

Park B: In another section of the estate, three survey markers are located at *P(1,0)m, Q(0,1)m,* and *R(3,4)m*.

A communication booster has to be installed at a distance equidistant from the fixed points of these circular parks. For proper inspection by the construction team, this distance must not exceed 1m. The construction company noted that a park – whose distance from the fixed point to the boundary will be larger – is to be used by motorists while the other by the cyclists.

After designing the parks, the company plans to construct an underground utility corridor (for water pipes, electrical cables, and communication lines) that follows the straight line joining two inspection chambers located at *S(1,-2,1)m* and *T(-1,3,4)m.*

Before excavation begins, the site engineer must verify whether an existing underground inspection point *P(0,1,2)m* is safely positioned relative to the proposed utility corridor. To determine whether relocation is necessary, the engineer calculates the perpendicular distance from the inspection point *P(0,1,2)m* to the line joining *S(1,-2,1)m* and *T(-1,3,4)m*. The underground utility corridor will be approved for construction if this perpendicular distance does not exceed 2m.

Task

(a) Help and suggest whether there will be proper inspection of the circular parks by the construction team.

(b) Help the team to assign the park to be used by the motorists and cyclists.

(c) Check whether the underground utility corridor will be approved for construction.

SECTION B

PART I

Respond to **one** *item in this* Part.

ITEM 2

A manufacturing company is preparing to launch a new production line for eco-friendly construction materials. To ensure efficient production and sustainable growth, the company's engineering, quality control, and finance departments use mathematical models to make informed decisions.

First, the quality control department models the relationship between the machine's operating efficiency and a calibration constant k. To guarantee that the machine operates at a single critical setting without fluctuations, the performance equation must have repeated roots. The model is given by $kx^2 - 28x + 49 = 0$. The engineers are required to determine the value of *k* that gives repeated roots and then find the repeated operating setting *x*. For limited fluctuations in machine operations, the operating setting *x* must be less than 5.

Next, the engineering design department studies how material expansion behaves under varying production conditions. The behavior is represented by the expression $(2 - x)^5$. The engineers are required to expand this expression in ascending powers of *x*. They then use the expansion to estimate the production scaling factor by evaluating $(1.25)^5$ to 3 decimal places. For production to be quick, the value obtained after evaluating $(1.25)^5$ should be greater than the value obtained at $x = 3.25$.

Finally, the production planning department forecasts future production using a geometric progression. The number of product batches manufactured each production cycle forms a GP. Historical records show that the sum of the first two production cycles is 9 batches, while the sum of the first four production cycles is 45 batches. Using this information, the production manager must determine the two possible values of the fifth production cycle to prepare raw materials and labor requirements. The raw materials and labor required will be approved only if the fifth production cycle at least generates a positive batch.

Task

(a) Help to assess whether there be limited fluctuations in machine operations.

(b) Help to find out whether production will be quick or not.

(c) Will the raw materials and labor requirements be approved?

ITEM 3

During the development of a new smart production system, the company's engineering team carries out three mathematical investigations.

The production manager is planning the manufacture of three different components: Component X, Component Y, and Component Z. The number of units produced, represented by x, y, and z, must satisfy information obtained from three production records. The X component costs Shs 2000, Y component costs Shs 3500 and Z component costs Shs 4000. The company has Shs 50,000 available for production of the components.

The first record shows that the total number of components produced is 9 units. The second record shows that twice component X plus three times component Y minus component Z gives a total of 9 units. The third record shows that three times component X minus component Y plus twice component Z gives a total value of 11 units.

The company's research department develops a monitoring device that represents certain electrical signals using complex numbers. During testing, a signal value is recorded as: $5+12i$. The engineers need to determine the complex signal value whose square produces this recorded signal. For signal testing to be effective, the magnitude of this complex signal value should be less than 4 units.

A computer algorithm used by the company processes data using logarithmic relationships. During calibration, the engineers need to find the input value x that satisfies the system condition involving logarithms with different bases. This relationship is given by the equation $\log_3 x + \log_x 9 = 3$. The calibration is satisfied if the smallest input value lies between 1 and 4.

Task

(a) With evidence, find out whether the company is able to produce the required, X, Y and Z components.

(b) Help to assess if the signal testing is effective.

(c) Will the calibration be satisfied?

PART II

Respond to **one** *item in this* Part*:*

ITEM 4

A commercial poultry farm is expanding its operations and the farm manager has hired you as a consultant to analyze different aspects of the project using calculus.

Chicken Growth Model:

The average weight *y* (in kilograms) of a selected broiler chicken changes with time *x* (in weeks) according to the differential equation $(x^2 + 1)\frac{dy}{dx} + 2xy = x$ where the chick has an initial weight of 1 kg at the beginning of the observation period, that is, y (0) = 1. The farmer is only allowed to sell any of these birds if the weight in 4 weeks is greater than or equal to 2kg.

Feed Storage Construction:

To support the increasing flock, the farm plans to fence a feed storage section whose upper boundary is represented by the curve $f(x) = 5 + 4x - x^2$. A sketch map of this upper boundary must be made for easy planning. Given that 4 birds occupy 1 square unit of space and the farmer has a total of 120 birds.

Water Tank Expansion:

The poultry farm uses a spherical water tank of radius 3cm to supply drinking water to the birds. Due to thermal expansion during the day, the surface area of the tank increases at a rate of 0.4cm^2/s. The rate of volume obtained is proportional to one bird. The farmer claims that the total number of birds will consume water at a rate of 72cm^3/s.

Task

(a) Find out whether the farmer will be allowed to sell the birds.

(b) Will the storage space be enough for the available number of birds?

(c) Verify whether the farmer's claim is true.

ITEM 5

A metal manufacturing company produces high-quality circular metal plates for use in heavy machinery. During production, engineers must estimate the amount of metal required for each plate, model equipment performance, and predict how the temperature of the metal changes after it leaves the furnace. As part of a quality control assessment, a trainee engineer is assigned the following tasks:

Material Estimation:

A cross-section of a circular metal plate is represented by the curve $y=\sqrt{1-x^2}$ for $0 < x<1$. To estimate the amount of metal required for one-quarter of the circular plate, the engineers need to find the cross-sectional area of y. For proper material estimation, the cross-sectional area should not be more 1 square unit.

Equipment Calibration:

During the rolling process, the efficiency factor of a machine is modeled by the function $(1-x)^{-2}$, where (x) is a small adjustment parameter. To simplify calculations for small values of (x), engineers need to find the Maclaurin's series of the efficient factor up to a term in x^3. The expansion represents the cost involved in the rolling process, given that each x unit costs Shs 100. The rolling process will be efficient if the total cost does not exceed Shs 2,000,000.

Cooling of the Metal:

After being removed from the furnace, a heated metal component begins to cool according to Newton's Law of Cooling, where the rate of cooling is proportional to the temperature difference between the metal and its surroundings. The metal leaves the furnace at a temperature of 120^0C and is placed in a workshop maintained at 20^0 C. After 6 minutes, its temperature is measured to be 95^0C. The engineers need to find the temperature after 15 minutes. The metal produced will be strong enough if its temperature at this time is above 60^0C.

Task

(a) Help and find out whether there is proper material estimation.

(b) Assess the effectiveness of the rolling process.

(c) With evidence, confirm that the metal produced will be strong.

END